

Problem Set 3

Due on Canvas at noon Friday October 9th

Instructions: No partial credit will be awarded. All answers must be entered in the Problem Set 3 Quiz on Canvas. Note that it is not possible to edit your responses on Canvas after submitting. **Best practice is to complete the problem set on paper and then transfer solutions to Canvas.** You may collaborate on this problem set with other students in your learning team, but each student must submit his or her own solutions.

Grading: Grading is done on a ten point scale across all problems in the problem set. **Note that after submitting on canvas, you will see a score, but that does not include any points for any qualitative questions.** Grades will be finalized a few days after the due date.

Material covered: This problem set covers lectures after 8 and including game theory.

Problem 1. Price Discrimination [3 points]

You have been hired by Kabletown, a small local cable television provider in Mountainville, as a consultant to advise them on pricing strategies. Due to the rough terrain and perpetual fog, no satellite television company can serve customers in Mountainville. As a result, Kabletown is the sole provider of cable television service in Mountainville.

You find out that there are two types of customers for Kabletown's services: **residential households** and **bars**. Residential household demand is given by

$$Q_H = 200 - 4P,$$

and Bar demand is given by

$$Q_B = 100 - P.$$

Kabletown's total cost function is

$$C(Q) = 6Q,$$

where Q is the total number of customers that subscribe to Kabletown's services. (Notice that there is no fixed cost here.)

Let us first assume that Kabletown cannot monitor if it is a household or bar that subscribes, and thus can only charge a uniform price in this market for cable television service.

- a. [0.5 pt] What is Kabletown's optimal uniform price?

Total demand is $Q = Q_H + Q_B = 300 - 5P$. Then inverse demand is $P = 60 - Q/5$; so $MR = 60 - 2Q/5$.

We also know $MC = 6$.

So $MR = MC$: $60 - 2Q/5 = 6$ yields $Q = 135$. Then the optimal uniform price is $P = 60 - 135/5 = 33$.

[In this case, the optimal profit = $(33 - 6) \times 135 = 3645$.]

You propose that Kabletown develop a new technology that allows them to monitor what type of customer subscribes to their service. This would also allow them to charge different prices to different types of customer.

- b. [0.5 pt] If Kabletown had access to this technology, what would the profit maximizing price be in the market for residential households?

Inverse demand is $P = 50 - Q_H/4$, so $MR = 50 - Q_H/2$. Then $MR = MC$ implies $50 - Q_H/2 = 6$. So

$Q_H = 88$, and $P_H = 28$.

- c. [0.5 pt] What would the profit maximizing price be in the market for bars?

Inverse demand is $P = 100 - Q_B$, so $MR = 100 - 2Q_B$. Then $MR = MC$ implies $100 - 2Q_B = 6$. So $Q_B = 47$, and $P_B = 53$.

- d. [0.5 pt] How much would Kabletown be willing to pay for this new technology?

Total profit with price discrimination would be:

$(28 - 6) \times 88 + (53 - 6) \times 47 = 1936 + 2209 = 4145$

So Kabletown's profit would increase from 3645 to 4145. That is, Kabletown would be willing to pay up to \$500 for this new technology.

- e. [1 pt] How much would residential households benefit or suffer from this new technology?

Under uniform price 33, consumer surplus in the residential market was $0.5 \times 17 \times 68 = 578$.
 New consumer surplus in the residential market would be: $0.5 \times 22 \times 88 = 968$.
 Therefore, residential customers would benefit $968 - 578 = 390$.

Problem 2. Versioning [3points]

The high-tech company Pear Inc. has developed an exciting new product: the Myphone with a large interactive screen and many customizable features. There are two possible versions of the new product: a regular version which costs **\$200** to produce, and a slightly higher-end version with more features called "Myphone Super" which costs **\$350** to produce. Pear Inc. is considering whether to supply one version of the product only or both versions and at what price(s). Pear's marketing departments suggests that consumers can be categorized as loyalists and casual users. Each consumer would consider at most one phone.

Consumer Type	Number of consumers	Willingness to pay for Myphone	Willingness to pay for Myphone Super
Casual	10 million	500	600
Loyalist	1 million	800	1100

- a. [2 pts] What would be Pear Inc.'s optimal selling and pricing strategy?

Strategy 1: Pear Inc. can choose to sell only Myphone Super

Then, should sell at \$600. This gives profit of $(600 - 350) \times 11 \text{ million} = 2.75 \text{ billion}$.

Strategy 2: Pear Inc. can choose to sell only Myphone.

Then should sell at \$500. This gives profit of $(500 - 200) \times 11 \text{ million} = 3.3 \text{ billion}$.

Strategy 3: Pear Inc. can choose to sell both versions.

(In this case, optimal strategy will be to sell Myphone at \$500 and Myphone Super at 800. To get the price of 800, notice that we need to set price of Myphone Super such that $1100 - p \geq 800 - 500$ (which is the surplus a loyalist can get from purchasing the regular version). Hence, we must have $p \leq 800$.)

This gives profit of $(500 - 200) \times 10 \text{ million} + (800 - 350) \times 1 \text{ million} = 3.45 \text{ billion}$.

Therefore, strategy 3 is the best, i.e., Pear Inc. should sell both versions: the regular version at \$500, and the Super version at \$800.

- b. [1pt] How would your answer change, if at all, if the causal segment were willing to pay \$675 for the Myphone Super?

Notice nothing changes in Strategy 2 & 3 above.

But now Strategy 1 gives a higher profit: $(675 - 350) \times 11 \text{ million} = 3.575 \text{ billion}$.

So now Pear Inc should sell Myphone Super only at \$675.

Problem 3. A game of launching new games [4 points]

Jokx Toys and Games is considering introducing a new game called *Oligopoly*, which will teach children from ages 10 to 15 the basic principles of imperfect competition. Jokx must decide very soon whether to introduce this game, and if it decides to go ahead, it will have to spend **\$4 million** to complete design of the game, advertise it, and set up production. In addition, it will cost Jokx **\$5** per unit to produce the game.

Jokx's competitor, Beljeau, is also considering the introduction of an economic education game called *Reaganomics*, that will compete directly with *Oligopoly*. Beljeau faces higher fixed costs than Jokx, i.e., it will cost **\$8 million** to start the project. But Beljeau will have the same per-unit production cost **\$5**.

From market research, both companies learn that there will be **2 million** potential customers, and each customer will buy one game if the price does not exceed their willingness-to-pay of **\$15**. If both games are introduced, competition will force the two companies to charge **\$12**, and they will split the market equally. (The two games are differentiated, and so the price competition will not be very fierce.) If only one company introduces its game, it will be a monopolist in the market.

The need to get these games out in time for Christmas means that each company cannot wait to see what their competitor does before deciding whether to go ahead: they must make decisions **simultaneously**. As an analyst, you are making a prediction about how many economics education games will be introduced into the market.

[Hint: A monopoly firm will charge a price at customers' willingness to pay, and all customers will buy at that price. Each firm, if it launches its new game, faces both a fixed cost and the per-unit production cost.]

- a. [1 pt] Which firm(s) have a dominant strategy?

Jokx has a dominant strategy which is "Launch Oligopoly".

The game matrix is as follows:

		Beljeau	
		Launch Reaganomics	Not Launch
Jokx	Launch Oligopoly	3, -1	16, 0
	Not launch	0, 12	0, 0

The calculation details are as follows:

Since each customer's willingness to pay for a game is \$15, the monopolist will charge at \$15. So Jokx's profit = $(15-5) \times 2m - 4m = 16m$, if it is the only producer.

Similarly, Beljeau's profit = $(15-5) \times 2m - 8m = 12m$, if it is the only producer.

If both introduce the game, the price will be \$12 and the two companies split the market equally, so

Jokx's profits = $(12-5) \times 1m - 4m = 3m$

Beljeau's profit= $(12-5) \times 1\text{m} - 8\text{m} = -1\text{m}$

- b. [1 pt] If you use Nash equilibrium (NE) to predict the market outcome, what is your prediction?

Given Jokx will adopt its dominant strategy Launch, Beljeau won't launch.
So Nash Equilibrium is (Launch Oligopoly, Not Launch)

Now suppose that besides the original two options of **"Launch"** and **"Not Launch"**, Beljeau has a **third** option: **launch Reaganomics aggressively** with high advertising intensity which would mean a **\$8.5 million** investment to start the project (which is 0.5 million higher than the previous investment needed for a normal launch). If Beljeau chooses this aggressive launch option, then it will have **70%** of the market share when both companies introduce their games. (The market price, if both companies launch, will be still \$12.)

- c. [1 pt] What is your prediction of the new game?

NE in the new game is (Launch Oligopoly, Aggressive Launch).

This can be seen from the matrix below:

		Beljeau		
Jokx		Launch Reaganomics	Aggressive Launch	Not Launch
	Launch Oligopoly	3, -1	0.2, 1.3	16, 0
	Not launch	0, 12	0, 11.5	0, 0

- d. [1 pt] Suppose now Beljeau can move first, what's your prediction of this sequential-move game?

By drawing a game tree and using backward induction, you can see the outcome is the same as in the simultaneous-move game, i.e., Beljeau will launch aggressively and Jokx will also choose to launch.